Completed TRIPOD+AI checklist

Manuscript. Feature Vectors and Narrative Embeddings for Predicting a Treatment Switching Proxy for Treatment Resistant Depression: Retrospective Cohort Study

Study type. Development of a multivariable prediction model with internal validation by a single random split. There is no external validation and no evaluation of a previously developed model, so items marked *E* (evaluation only) are recorded as not applicable unless the internal-validation split addresses them.

Key. D = development; E = evaluation; D;E = both. "Reported in" refers to sections of the main manuscript unless prefixed *Supplement*.

Title and abstract

Item	D/E	Checklist item	Reported in
1	D;E	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted	Title page. The title names the prediction task, the outcome (a treatment-switching proxy for treatment-resistant depression), and the study design; the target population (patients with depression at their index antidepressant prescription in one community health system) is specified in the Abstract and in Methods, *Participants and Prediction Time*
2	D;E	See TRIPOD+AI for Abstracts checklist	Abstract (structured: Background, Objective, Methods, Results, Conclusions)

Introduction

Item	D/E	Checklist item	Reported in
3a	D;E	Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing or evaluating the prediction model, including references to existing models	Introduction, paragraphs 1–3. Prognostic. Existing models cited [4-8,12-15]
3b	D;E	Describe the target population and the intended purpose of the prediction model in the context of the care pathway, including its intended users	Introduction, paragraphs 1 and 5 (identifying, at the index antidepressant prescription, patients likely to have a difficult treatment course, to inform monitoring and follow-up); Discussion, *Implications and Next Steps* (the results do not support clinical deployment; outcome validation, temporal and external evaluation, and prespecified thresholds come first)
3c	D;E	Describe any known health inequalities between sociodemographic groups	Methods, *Outcome*: access to depression treatment differs across US racial and ethnic groups, so the label can record unequal access as clinical signal (with Supplement M3). Discussion, *Clinical Interpretation and Limitations*, paragraph 3: the outcome depends on access to treatment, which is documented to differ across racial and ethnic groups in US depression care [28,29]. Supplement M3 states that switching depends on continuity and access, which differ across demographic groups. Addressed empirically in Results, *Subgroup Performance*, Supplement S9 (subgroup discrimination and calibration) and Supplement S11 (subgroup outcome frequency)

Item	D/E	Checklist item	Reported in
4	D;E	Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both)	Introduction, final paragraph (the primary question and three secondary questions). Development with internal validation

Methods

Item	D/E	Checklist item	Reported in
5a	D;E	Describe the sources of data separately for the development and evaluation datasets, the rationale for using these data, and representativeness of the data	Methods, *Study Design and Data Source*; Supplement M1. A frozen, de-identified Epic EHR extract from one community health system; development and evaluation samples are a single random partition of that one extract, whose versions and dates are stated. Representativeness: the sample is enriched for depression and does not represent health-system prevalence (Methods, *Study Design and Data Source*; Supplement M1)
5b	D;E	Specify the dates of the collected participant data, including start and end of participant accrual; and, if applicable, end of follow-up	Methods, *Study Design and Data Source* (index dates 2013–2025; extracted June 29, 2026); Methods, *Participants and Prediction Time* (730-day lookback and at least 365 days of follow-up). Figure 1 shows the lookback window, the index, and the outcome window on one axis

Item	D/E	Checklist item	Reported in
6a	D;E	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including the number and location of centres	Methods, *Study Design and Data Source*, and Supplement M1. Single centre: Saint Francis Health System, Tulsa, Oklahoma, United States — one community health system, covering inpatient, outpatient, observation, and emergency encounters within one administrative division, entered through routine antidepressant prescribing rather than psychiatric specialty referral
6b	D;E	Describe the eligibility criteria for study participants	Methods, *Participants and Prediction Time*; Supplement M1 (diagnosis code lists) and M2, Table M1 (six hierarchical filters)
6c	D;E	Give details of any treatments received, and how they were handled during model development or evaluation, if relevant	Methods, *Participants and Prediction Time* (the index is the earliest antidepressant prescription on or after the first documented depression diagnosis; prior antidepressant use is a predictor, not an exclusion); Methods, *Patient Representations and Missing Data* (prior trial counts at 42 days of continuous exposure, which does not establish adequate dose); Methods, *Supporting Analyses* (prior treatment exposure is one of the 6 permuted concepts; Supplement M11 defines it as pre-index exposure, distinct from the post-index agents that construct the outcome). Results, *Clinical Contributions and Encoder Robustness*, Table 3; Supplement S12.3

Item	D/E	Checklist item	Reported in
7	D;E	Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups	Methods, *Patient Representations and Missing Data*; Methods, *Model Development and Evaluation* (scaling and category encoding fitted within each training fold); Supplement M4, M9 and S10. Pre-processing is identical for all patients. Differences in data completeness: Supplement M6 (religion missingness by age band; body mass index missingness by outcome status)
8a	D;E	Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome, and whether the method of outcome assessment is consistent across sociodemographic groups	Methods, *Outcome*. At least 3 distinct antidepressant treatments within 365 days, the index agent counting as the first; augmentation without replacement and restarts do not count. Supplied as a fixed upstream label [20]. The label is computed algorithmically from prescription records and is applied identically to every patient; Supplement M3 states that what it captures depends on access, which differs across demographic groups. Limits as a proxy: Methods, *Outcome*; Supplement M3; Discussion, *Clinical Interpretation and Limitations*, paragraph 1
8b	D;E	If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors	Not applicable. The outcome is derived algorithmically from structured prescription records; no human assessor interprets it
8c	D;E	Report any actions to blind assessment of the outcome to be predicted	Methods, *Outcome*; Supplement M3. The label was generated in a separate upstream data-preparation pipeline, independently of the predictors, and supplied as a fixed label

Item	D/E	Checklist item	Reported in
9a	D	Describe the choice of initial predictors and any pre-selection of predictors before model building	Methods, *Patient Representations and Missing Data* (predictor domains selected before model fitting or examination of outcome associations [4-7]); Supplement M4 (no univariate screening or automated selection; Tables M2–M4). Three vital-sign fields were removed from FEATURE for missingness (Supplement M4, M6)
9b	D;E	Clearly define all predictors, including how and when they were measured (and any actions to blind assessment of predictors for the outcome and other predictors)	Methods, *Participants and Prediction Time* (clinical content restricted to the 730-day window ending on the index date); Supplement M4, Tables M2–M4; Supplement S10, Table S13 (field-level crosswalk). No post-index information enters any predictor
9c	D;E	If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors	Not applicable. All predictors are derived deterministically from structured EHR fields; the narratives are written by fixed rules, with no generative model and no human judgement (Methods, *Patient Representations and Missing Data*; Supplement M5)
10	D;E	Explain how the study size was arrived at (separately for development and evaluation), and justify that the study size was sufficient to answer the research question. Include details of any sample size calculation	Supplement M7. No formal sample-size calculation; cohort size fixed by eligibility. Events per variable reported per representation and per encoder against the conventional threshold of 10 (Supplement M7, Table S15)

Item	D/E	Checklist item	Reported in
11	D;E	Describe how missing data were handled. Provide reasons for omitting any data	Methods, *Patient Representations and Missing Data*; Supplement M6. No statistical imputation. Three vital-sign fields removed from FEATURE because their missingness is associated with the outcome; missing categorical values retained as explicit levels; narrative missing-value rules in Supplement S10. Two sensitivity analyses (vital signs retained with missingness indicators and within-fold median imputation; religion removed from both pipelines) are described in Supplement M6, with quantitative results available from the corresponding author
12a	D	Describe how the data were used in the analysis, including whether the data were partitioned, considering any sample size requirements	Methods, *Model Development and Evaluation* (one stratified 80:20 split shared by every analysis); Supplement M8 (the retrieval index holds training patients only)
12b	D	Depending on the type of model, describe how predictors were handled in the analyses (functional form, rescaling, transformation, or any standardisation)	Methods, *Patient Representations and Missing Data* (continuous variables standardized, categorical one-hot encoded, binary coded 0 or 1); Supplement M9 (the embedding enters as a single numeric block)

Item	D/E	Checklist item	Reported in
12c	D	Specify the type of model, rationale, all model-building steps, including any hyperparameter tuning, and method for internal validation	Methods, *Model Development and Evaluation* (4 classifiers per representation; five-fold cross-validated grid search optimizing ROC AUC; refit on all training patients; the same grid for both representations, with hyperparameters selected separately for each); Supplement M9. Retrieval: every neighborhood size from 1 to 34,063 was scored and each arm is reported as a curve over every k. Apart from those curves, every performance value reported for a retrieval arm — ROC curves, confusion matrices, maxima, and the AUPRC, Brier and calibration summaries — is at that arm's best k, except where a fixed k is named; that k was chosen on the test patients, so these values are optimistic (Methods, *Supporting Analyses*; Supplement M10). Internal validation is the held-out 20% split
12d	D;E	Describe if and how any heterogeneity in estimates of model parameter values and model performance was handled and quantified across clusters	Not applicable. Data come from a single source with no modelled cluster structure; no clustered or multi-centre analysis was performed

Item	D/E	Checklist item	Reported in
12e	D;E	Specify all measures and plots used (and their rationale) to evaluate model performance and, if relevant, to compare multiple models	Methods, *Model Development and Evaluation* (ROC AUC, AUPRC, Brier score, calibration; bootstrap CIs; paired bootstrap for differences between models scored on the same patients; the best-model comparison is post hoc and no equivalence margin was prespecified); Supplement M13. Precision–recall under class imbalance: Supplement S2, including the 3 retrieval arms at their best k (Table S1, Figure S4C–E). Calibration: Supplement S3, including the retrieval arms (Tables S3–S4, Figure S5C–E), and S9. Random retrieval with uniform weights, repeated in 1,000 draws, is the floor for retrieval (Figure 4)
12f	E	Describe any model updating arising from the model evaluation, either overall or for particular sociodemographic groups or settings	Not applicable. No recalibration or model updating was performed after evaluation
12g	E	For model evaluation, describe how the model predictions were calculated	Methods, *Model Development and Evaluation* (classifier predicted probabilities from the refitted pipeline); Methods, *Supporting Analyses*, Equations 1–2 (the neighbor-retrieval risk score in closed form); Supplement M10; Data Availability (analysis code)
13	D;E	If class imbalance methods were used, state why and how this was done, and any subsequent methods to recalibrate the model or the model predictions	No class-imbalance method was used: no resampling, class weighting, or recalibration, stated in Methods, *Model Development and Evaluation*. The 17.5% (95% CI 17.2–17.9) positive rate was preserved in both partitions by the stratified split (Methods, *Model Development and Evaluation*; Supplement M8). Consequences for interpretation: precision–recall analysis (Supplement S2) and calibration (Supplement S3)

Item	D/E	Checklist item	Reported in
14	D;E	Describe any approaches that were used to address model fairness and their rationale	No fairness assessment was performed, and the manuscript says so. Methods, *Supporting Analyses*: race/ethnicity and social determinants are among the permuted concepts, a test of direct input reliance that does not establish fairness (Supplement M11). Subgroup performance: Results, *Subgroup Performance* (Figure 5) reports ROC AUC contrasts; Supplement S9 (Tables S10–S12, Figure S13) adds Brier score, logistic calibration slope and mean risk difference, each with a bootstrap 95% CI, across sex, recorded race, age, marital status, smoking, religion, MDD recurrence and severity, for 4 FEATURE classifiers, 4 EMBEDDED classifiers and 2 retrieval configurations at their best k, with strata below 20 in the smaller outcome class declared not estimable. 240 contrasts with bootstrap P values adjusted by Benjamini–Hochberg; 58 exclude zero unadjusted and 23 survive, 19 of them involving recurrence. All 10 White-minus-non-White contrasts are positive, from +0.005 (95% CI −0.031 to +0.042) to +0.046 (95% CI +0.006 to +0.087); 5 exclude zero unadjusted and none survive (minimum adjusted P = .15); in Supplement S9, calibration slopes in White patients and in other recorded racial categories are 0.97 (95% CI 0.87–1.08) and 0.95 (95% CI 0.74–1.16) for embedded logistic regression and 1.28 (95% CI 1.14–1.44) and 1.07 (95% CI 0.75–1.36) for feature-vector XGBoost, with 1,181 against 302 events. Discussion, *Clinical Interpretation and Limitations*, paragraph 3, states that this does not establish fairness and why

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15	D	Specify the output of the prediction model. Provide details and rationale for any classification and how the thresholds were identified	Methods, *Model Development and Evaluation*. The model outputs a predicted probability of the TRD proxy. Sensitivity and specificity at the test-selected Youden J threshold are reported only as descriptive analyses (Supplement M13, S12.1, Figures S14–S15), because selecting and evaluating a threshold in the same patients is optimistic. No deployable threshold is proposed; Discussion, *Implications and Next Steps* calls for prespecified thresholds
16	D;E	Identify any differences between the development and evaluation data in healthcare setting, eligibility criteria, outcome, and predictors	None: both partitions are one random stratified split of one cohort, with identical setting, eligibility, outcome, and predictors (Methods, *Model Development and Evaluation*; Supplement M8). Balance: Results, *Cohort Characteristics* (maximum absolute standardized mean difference 0.036); Supplement S8, Table S9
17	D;E	Name the institutional research board or ethics committee that approved the study and describe the participant-informed consent or the ethics committee waiver of informed consent	Methods, *Ethical Considerations*. Secondary analysis of de-identified records; no institutional review board review or consent waiver was obtained because the study was considered not to involve human participants

Open science

Item	D/E	Checklist item	Reported in
18a	D;E	Give the source of funding and the role of the funders for the present study	Funding. The William K. Warren Foundation, with no other funding agency; the funder had no role in design, analysis, interpretation, writing, or the decision to submit
18b	D;E	Declare any conflicts of interest and financial disclosures for all authors	Conflicts of Interest. None declared
18c	D;E	Indicate where the study protocol can be accessed or state that a protocol was not prepared	Protocol and Registration. No study protocol was prepared
18d	D;E	Provide registration information for the study, including register name and registration number, or state that the study was not registered	Protocol and Registration. The study was not registered
18e	D;E	Provide details of the availability of the study data	Data Availability. The EHR data cannot be shared publicly
18f	D;E	Provide details of the availability of the analytical code	Data Availability. Analysis code is public at https://github.com/Pirate-Hunter-Zoro/Atlas/tree/da44a79a7a68f7e57de2e2ba2e27a59835de9530/projects/TRD-EHR; software is named in Methods, *Supporting Analyses* [22,26,27]

Patient and public involvement

Item	D/E	Checklist item	Reported in
19	D;E	Provide details of any patient and public involvement during the design, conduct, reporting, interpretation, or dissemination of the study or state no involvement	Methods, *Ethical Considerations*. No patients or members of the public participated

Results

Item	D/E	Checklist item	Reported in
20a	D;E	Describe the flow of participants through the study, including the number of participants with and without the outcome and, if applicable, a summary of the follow-up time	Results, *Cohort Characteristics*; Supplement M2, Table M1 (501,718 → 42,579; 7,455 outcome-positive, 17.5%, 95% CI 17.2–17.9). At least 365 days of follow-up is imposed by eligibility (Methods, *Participants and Prediction Time*)
20b	D;E	Report the characteristics overall and, where applicable, for each data source or setting, including the key dates, key predictors, treatments received, sample size, number of outcome events, follow-up time, and amount of missing data. Report any differences across key demographic groups	Results, *Cohort Characteristics*, Table 1 (overall and by outcome status, with standardized mean differences); Supplement S11, Table S14 (expanded characteristics; outcome frequency by sex, age and social-determinant strata); Supplement M6 (amount and pattern of missingness)
20c	E	For model evaluation, show a comparison with the development data of the distribution of important predictors	Results, *Cohort Characteristics*; Supplement S8, Table S9 (training against test on the Table 1 characteristics). Maximum absolute standardized mean difference 0.036 over 100 predictor rows
21	D;E	Specify the number of participants and outcome events in each analysis	Methods, *Model Development and Evaluation*; Supplement M7, M8. Training 34,063 (5,964 events); test 8,516 (1,491 events). Per-subgroup counts and events: Supplement S9, Tables S10–S11
22	D	Provide details of the full prediction model to allow predictions in new individuals and to enable third-party evaluation and implementation, including any restrictions to access or re-use	**Partially reported.** The analysis code, including model specification and fitting, is public (Data Availability), and the retrieval predictor is given in closed form (Methods, *Supporting Analyses*, Equations 1–2). Fitted model objects are not distributed because they derive from non-shareable patient data (Data Availability)

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23a	D;E	Report model performance estimates with confidence intervals, including for any key subgroups. Consider plots to aid presentation	Results, *Discrimination and Calibration* (Table 2, Figure 2); *Clinical Contributions and Encoder Robustness* (Table 3, Figure 3); *Nearest-Neighbor Retrieval* and *Nearest-Neighbor Retrieval Across Encoders* (Figure 4, all 4 encoders, every k from 1 to 34,063 with bootstrap 95% bands for the nearest-neighbor arms and the random floor's 2.5th–97.5th percentile band, and each arm's best k with its 95% CI); Supplement S2–S3 (Tables S1–S4, every classifier and each retrieval arm at its best k with a bootstrap 95% CI; the random arm's binned calibration slope and intercept are not estimable, Table S4), S6 (Table S7; Figures S8–S9, ROC curves and confusion matrices at best k for all four encoders; Figures S10–S11, logistic-regression dimensions against best k), S7 (Table S8), S12 (Table S15). Subgroup performance: Results, *Subgroup Performance* (Figure 5, every contrast surviving adjustment, with its 95% CI) and Supplement S9 (Tables S10–S12, Figure S13)
23b	D;E	If examined, report results of any heterogeneity in model performance across clusters	Not applicable. Not examined; see item 12d
24	E	Report the results from any model updating, including the updated model and subsequent performance	Not applicable. No model updating was performed; see item 12f

Discussion

Item	D/E	Checklist item	Reported in
25	D;E	Give an overall interpretation of the main results, including issues of fairness in the context of the objectives and previous studies	Discussion, *Principal Findings* and *Comparison With Prior Work* [12-15]; retrieval in *Interpreting Nearest-Neighbor Retrieval* [30], including that best k was chosen on the test patients [31]. Fairness: Discussion, *Clinical Interpretation and Limitations*, paragraph 3 [28,29]
26	D;E	Discuss any limitations of the study and their effects on any biases, statistical uncertainty, and generalizability	Discussion, *Clinical Interpretation and Limitations*. The main limitation is the target: switching can reflect tolerability, preference, clinician behavior and access as well as inadequate response, and the label was not validated against failed adequate trials or symptom trajectories [9-11]. Further: both representations encode the same hand-selected predictors, with the narrative a fixed template over them, so the comparison is between two encodings of curated fields and not evidence about an unselected record; selection and observation bias from the follow-up requirement, the exclusion of deceased patients and missed out-of-system care; the single-system, depression-enriched sample and random internal validation, which limit transportability and absolute risk; fairness not established, with small subgroup event counts and broad racial aggregation [28,29]; and concept permutation as sensitivity rather than causal contribution. Statistical uncertainty: Methods, *Model Development and Evaluation* (post hoc best-model comparison; no equivalence margin). Unequal field content between the pipelines: Methods, *Patient Representations and Missing Data*; Supplement S10

Item	D/E	Checklist item	Reported in
27a	D	Describe how poor quality or unavailable input data should be assessed and handled when implementing the prediction model	**Not reported.** Missing-data handling is specified for model development (Methods, *Patient Representations and Missing Data*) but no implementation-time guidance is given, consistent with the study not supporting deployment (Discussion, *Implications and Next Steps*)
27b	D	Specify whether users will be required to interact in the handling of the input data or use of the model, and what level of expertise is required of users	Not applicable. The results do not support clinical deployment (Discussion, *Implications and Next Steps*), and no user-facing implementation is proposed
27c	D;E	Discuss any next steps for future research, with a specific view to applicability and generalizability of the model	Discussion, *Implications and Next Steps*: validate the switching phenotype against treatment histories and symptom change; evaluate frozen pipelines in temporal and external cohorts; compare embeddings with compact psychiatric-history, medication and utilization baselines, reporting precision–recall, individual-level calibration, subgroup errors and net benefit at prespecified thresholds; and consider a more clinically informative target. Discussion, *Clinical Interpretation and Limitations*, paragraph 1 (variation across EHR definitions, with unstructured record elements [9-11])

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